COMPUTER AIDED DRAFTING SOFTWARE PROJECT

RCC COLUMN FOOTING LAYOUT MARKING BY USING AUTOCAD

PROJECT REPORT

CIVIL ENGINEERING – COMPUTER AIDED DRAFTING

1. ABSTRACT

This project presents the preparation of an RCC column footing layout marking using AutoCAD. The work is based on the given column footing layout plan and the separate column detailing schedule. The plan shows grid lines, column positions, footing locations and dimensional marking, while the column schedule gives column types, quantities, sizes, longitudinal reinforcement and tie details. AutoCAD is used to prepare a clear, accurate and well-organized engineering drawing for civil engineering drafting practice.

2. INTRODUCTION

A footing is the foundation element that transfers the load from a column to the supporting soil. Proper marking of column locations and footing positions is important for a coordinated structural layout. In this project, a grid-based RCC column footing layout is represented in plan, and the column types are detailed separately. The drawing is prepared in AutoCAD using standard drawing, modification, dimensioning and text commands.

3. AIM AND OBJECTIVES

Aim: To prepare an RCC column footing layout marking and column detailing drawing using AutoCAD based on the given dimensions and schedules.

Objectives:

- Understand grid-based column and footing layout marking.
- Locate columns accurately on grid intersections.
- Represent footing positions and column marks clearly.
- Prepare a separate column schedule/detailing drawing.
- Practice AutoCAD drawing and modification commands.
- Add dimensions, text and annotations clearly.
- Develop professional civil engineering drafting skills.

4. DRAWING DIMENSIONS

All dimensions are in millimetres. Horizontal grid spacing: 1–2 = 4075 mm and 2–3 = 3465 mm. Vertical grid spacing: A–B = 1900 mm, B–C = 1760 mm, C–D = 3770 mm and D–E = 5065 mm. Column sizes in the supplied schedule: C1–C4 = 230 × 450 mm; C5 = circular DIA 300 mm. The supplied plan includes column/footing marks such as C2/F2/P2, C3/F2/P2, C4/F3/P3, C1/F2/P2 and C5/F1/P1.

5. COLUMN SCHEDULE – SEPARATE

TYPE	NO.S	L (mm)	B (mm)	REINFORCEMENT	TIES
C1	1	230	450	Y16 – 8 NOS Y8 @ 200 C/C	1 TIE
C2	3	230	450	Y16 – 10 NOS Y8 @ 200 C/C	2 TIE
C3	3	230	450	Y20 – 6 NOS + Y16 – 2 NOS Y8 @ 200 C/C	1 TIE
C4	2	230	450	Y20 – 10 NOS Y8 @ 200 C/C	2 TIE
C5	2	DIA 300	---	Y16 – 8 NOS	1 TIE

Note: The schedule is transcribed from the supplied column detailing drawing for this academic project.

6. DRAWING 1 – COLUMN FOOTING LAYOUT PLAN

The plan shows grid lines 1–3 and A–E, footing/column locations, column identification and the given horizontal and vertical dimensions.

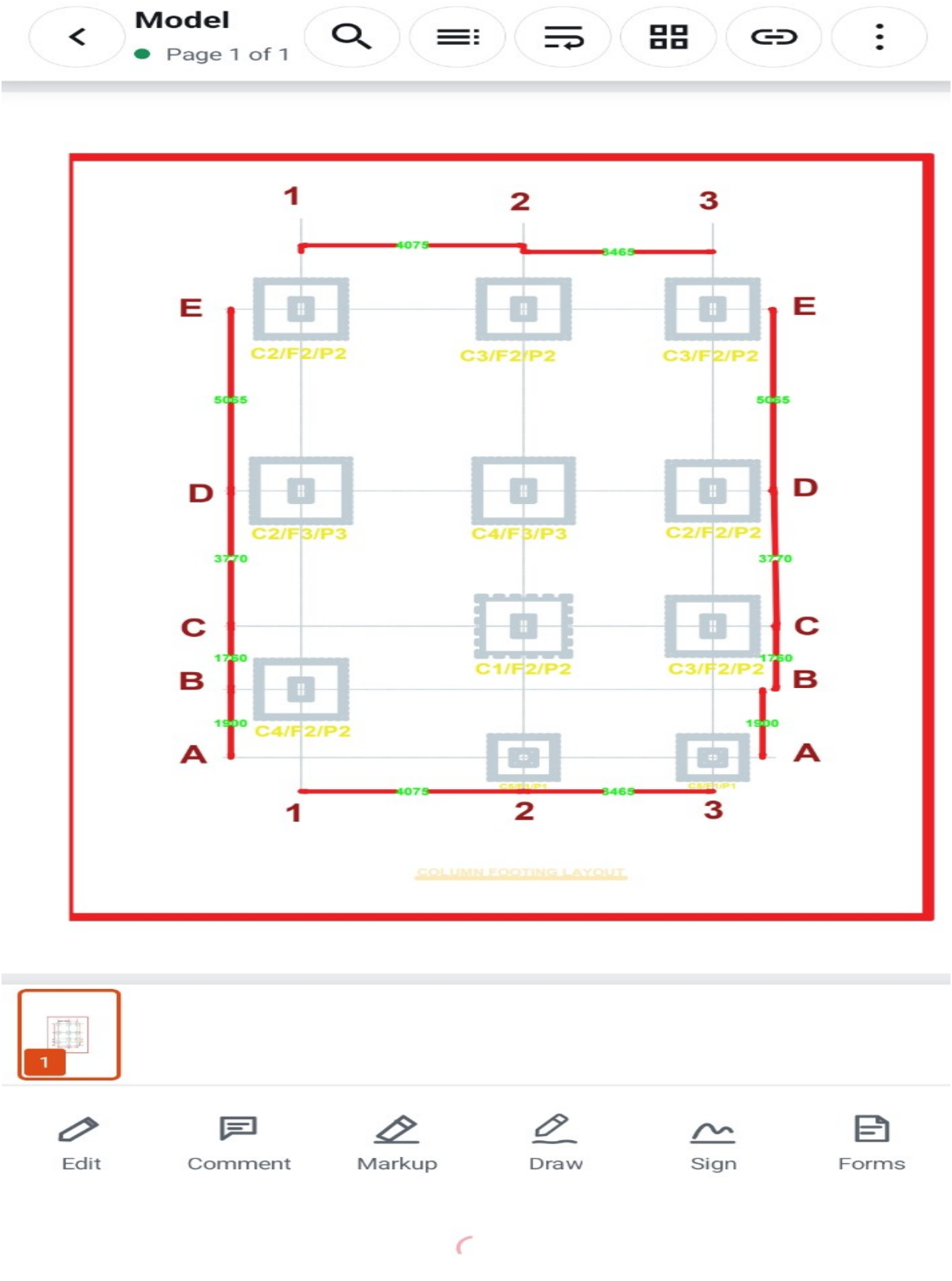
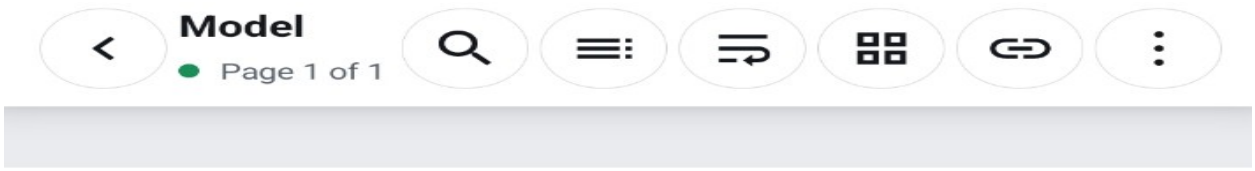


Figure 1: Supplied column footing layout plan

7. DRAWING 2 – COLUMN DETAILING

The separate detailing drawing records column types C1 to C5, number of columns, dimensions, longitudinal reinforcement and tie information.



TYPE	NO.S	COLUMN		TYP.C/S OF COLUMN	COLUMN TIES
		L	B		
C1	1	230	450	Y 16-8 NO.S Y 8-200 C/C	1 TIE
C2	3	230	450	Y 16-10NO.S Y 8-200 C/C	2 TIE
C3	3	230	450	Y 20-6NO.S +Y 16-2NO.S Y 8-200 C/C	1 TIE
C4	2	230	450	Y 20-10 NO.S Y 8-200 C/C	2 TIE
C5	2	DIA 300	---	Y 16- 8 NO.S	1 TIE

COLUMN DETAILING

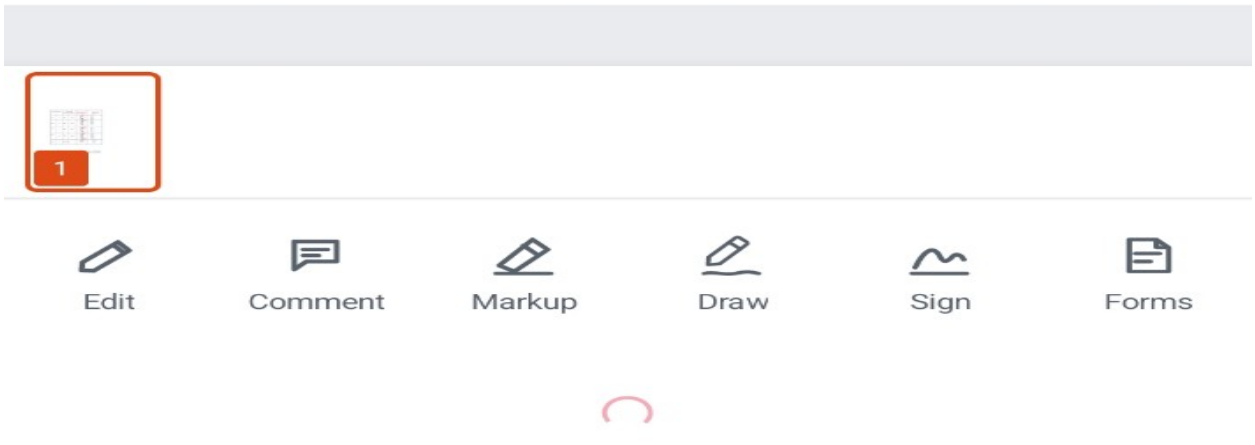


Figure 2: Supplied column detailing

8. TECHNOLOGY AND SOFTWARE USED

Software: Autodesk AutoCAD.
Technology: Computer Aided Drafting (CAD). AutoCAD is used for grid creation, column and footing representation, dimensions, annotations and detailing.

9. DRAWING TOOLS

LINE, PLINE, RECTANGLE, CIRCLE, OFFSET, ARRAY, HATCH and POLYGON are used for creating grid lines, outlines, column symbols, footing representation and repeated details.

10. MODIFICATION TOOLS

MOVE, COPY, MIRROR, TRIM, EXTEND, ROTATE, FILLET and ERASE are used to position, duplicate, modify and clean the drawing.

11. DIMENSIONS AND TEXT

Dimension commands are used for the given grid spacings: 4075, 3465, 1900, 1760, 3770 and 5065 mm. Text is used for grid labels, column/footing marks, titles and notes. The column schedule records 230 × 450 mm rectangular columns and DIA 300 mm circular columns.

12. APPLICATIONS

• RCC foundation layout drawings.
• Column and footing location marking.
• Structural drafting and documentation.
• Academic civil engineering CAD projects.
• Coordinated plan and detailing drawings.
• Digital engineering drawing preparation.

13. ADVANTAGES

• Accurate grid and column positioning.
• Clear footing representation.
• Easy editing of dimensions and text.
• Separate detailing for different column types.
• Easy storage, modification and reproduction.
• Professional presentation of engineering drawings.

14. RESULT

The RCC column footing layout marking was prepared in AutoCAD based on the supplied plan and column detailing drawings. The project includes the grid-based layout, dimensional marking, column identification and separate C1–C5 column schedule with the given reinforcement details.

15. CONCLUSION

The project demonstrates the use of AutoCAD for preparing an RCC column footing layout and column detailing drawing. The exercise develops understanding of grid marking, column scheduling, dimensioning, annotation and professional civil engineering CAD documentation.